

AI-Based Quality Risk Prediction Report

Date: 20-07-2026

Time: 01:07 PM

Prediction Summary

Prediction	FAIL
Failure Risk	87.83%
Risk Level	High Risk
Likely Defect	OD Out of Tolerance
Confidence	91.33%

Input Parameters

Parameter	Value
Material	PU
Braid	None
Machine	EXT-01
Target OD (mm)	12.0
Target ID (mm)	9.5
Wall Thickness (mm)	1.25
OD Tolerance (mm)	0.15
ID Tolerance (mm)	0.15
Wall Tolerance (mm)	0.08
Vacuum (bar)	-0.5833
Screw Speed (RPM)	49.74
Zone 2 Temp (°C)	224.97
Zone 3 Temp (°C)	226.37
Melt Pressure (bar)	160.77
Puller Speed (m/min)	15.026

Material Moisture (%)	0.0673
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AI Recommendations

- Increase Melt Pressure from 160.77 to 168.81 bar
- Decrease Target OD from 12.00 to 11.40 mm
- Increase Screw Speed from 49.74 to 52.23 RPM
- Decrease Zone 2 Temperature from 224.97 to 213.72 °C
- Increase Material Moisture from 0.07 to 0.08 %

Generated by: AI-Based Thermoplastic Extrusion Quality Prediction System